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GARDEN CITY UNIVERSITY

EMPHASIS ON LIFE

Established under Karnataka State Act 47 of 2013 and Approved by UGC, Govt of India

DR. APJ ABDUL KALAM SCHOOL OF ENGINEERING

BENGALURU, KARNATAKA, INDIA - 560049

Smart Helmet for Accident Prevention and Safety

Report Submitted to the Garden City University

as the requirement for the course

10ABTEC22758: PILOT PROJECT

Submitted by

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November 2025

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Bonafide Certificate

This is to certify that the report titled "**Smart Helmet for Accident Prevention and Safety**" submitted in partial fulfilment of the requirements for the award of the degree of **B. Tech. Computer Science & Engineering (Specialization)** to the Garden City University, is a bonafide record of the work done by **S Shalini(22BTDS174), Navaneeth Yadav (22BTDS119), Thanushree N S (22BTDS155)** and **Anushree V S (22BTDS145)** during the 7th semester of the academic year 2025-26, in the **Dr. APJ Abdul Kalam School of Engineering**, under my supervision. This report has not formed the basis for the award of any degree, diploma, associateship, fellowship or other similar title to any candidate of any University.

Signature of the Pilot Project Co-ordinator :

Name and Affiliation :

Date :

Review held on _____

Examiner 1 **Examiner 2**

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Declaration

We declare that the report titled "**Smart Helmet for Accident Prevention and Safety**" submitted by us is an original work done by us under the guidance of **(Pilot Project Guide name with Affiliation)** during the 7th semester of the academic year 2025-26, in the **Dr. APJ Abdul Kalam School of Engineering**. The work is original and wherever we have used materials from other sources, we have given due credit and cited them in the text of the thesis. This thesis has not formed the basis for the award of any degree, diploma, associate-ship, fellowship or other similar title to any candidate of any University.

Signature of the Candidates:

(S Shalini) (Navaneeth Yadav)

(Thanushree) (Anushree)

Date :

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Abstract

The ***Smart Helmet for Accident Prevention and Safety*** is an innovative pilot project designed to enhance motorcycle rider protection through intelligent sensing, real-time monitoring, and automated emergency response. Road accidents involving two-wheelers continue to be a major cause of fatalities, often due to delayed medical assistance and the absence of preventive safety measures. This project aims to address these challenges by integrating IoT-based technologies within a wearable helmet system.

The proposed prototype incorporates multiple sensors such as an accelerometer and gyroscope for crash detection, an alcohol sensor to monitor rider sobriety, and a helmet-wear detection module to ensure mandatory usage. When a high-impact collision is detected, the system immediately triggers an automated alert message containing the rider's GPS location, which is transmitted to emergency contacts via GSM. This rapid communication significantly reduces the response time during critical situations, potentially saving lives. Additionally, the alcohol detection mechanism interfaces with the vehicle's ignition unit, preventing the engine from starting if the rider is intoxicated, thereby reducing the risk of accidents caused by impaired driving.

The pilot project demonstrates how embedded systems, IoT communication, and safety sensors can be integrated seamlessly into a compact, user-friendly, and low-cost safety device. Through testing and evaluation, the prototype exhibits high reliability in accident identification and emergency communication. The Smart Helmet thus serves as an effective solution to enhance road safety, promote responsible riding behavior, and contribute to a more intelligent transportation ecosystem.Designed and developed an IoT-based accident detection system using accelerometer and gyroscope sensors.

- ✓ Integrated GSM/GPS modules for automated location-based emergency alerts.
- ✓ Implemented alcohol sensing with ignition-locking capability to prevent drunk driving.
- ✓ Developed a functional prototype helmet with real-time monitoring and safety automation features.
- ✓ Gained hands-on experience in embedded systems, microcontroller programming, and sensor calibration.
- ✓ Learned practical implementation of IoT communication using GSM and GPS modules.

1. INTRODUCTION

1.1 Introduction

The safety of two-wheeler riders has become a major concern due to increasing road accidents and delayed emergency response. A well-designed safety system ensures real-time monitoring, timely accident alerts, and preventive actions against drunk driving, ultimately safeguarding the rider's life. However, most existing systems rely on basic helmets, manual reporting, or disconnected technologies, resulting in delayed assistance, misuse of vehicles under alcohol influence, and lack of accurate location tracking.

This project introduces a **Smart Helmet Safety System**, an intelligent and automated solution that integrates alcohol detection, accident/fall detection, GPS tracking, and instant emergency alerts through Telegram. Unlike traditional helmets or standalone devices, this system uses an ESP32-based platform to centralize all safety functions, automatically restricts vehicle ignition when alcohol is detected, and immediately sends the rider's live location to emergency contacts in case of an accident. The Smart Helmet aims to provide a centralized, efficient, and real-time safety mechanism for riders, promoting responsible driving and saving lives through rapid intervention.

1.2 Purpose of Project

The primary purpose of developing the Smart Helmet is to enhance road safety by detecting accidents in real-time, monitoring the rider's alcohol level, and ensuring that the rider wears the helmet properly before the vehicle starts. This system aims to reduce road fatalities, ensure responsible driving behavior, and assist in quick rescue operations during emergencies.

The specific objectives of the Smart Helmet are:

1.2.1 Safety Enforcement

- Integrate a helmet detection sensor to ensure that the rider is wearing the helmet before starting the vehicle.
- Lock vehicle ignition using the L298N motor driver if the helmet is not detected, preventing unsafe riding.
- Monitor helmet status continuously to detect if the helmet is removed while riding.
- Provide visual (LED) and audible (buzzer) alerts when helmet is not worn properly.
- Reduce accident risks due to non-compliance with helmet safety regulations.

1.2.2. Accident Detection & Alert

- Utilize MPU6050 accelerometer and gyroscope to detect abnormal tilts, sudden jerks, or collisions.
- Define threshold limits to distinguish between normal riding vibrations and actual accident events.
- Automatically send real-time alerts with rider location using GPS data through Telegram or GSM to emergency contacts.
- Trigger buzzer and LED indicators to notify nearby people about the accident.
- Minimize response time for medical assistance, increasing the chances of saving lives.

1.2.3. Alcohol Detection

- Implement MQ3/MQ2 alcohol sensor to measure alcohol concentration in the rider's breath.

- Prevent vehicle ignition if alcohol level exceeds a predefined safe limit, enforcing sobriety.
- Generate instant alerts to connected contacts or mobile app when alcohol is detected.
- Log instances of alcohol detection for monitoring and reporting purposes.
- Enhance preventive safety by reducing drunk driving incidents.

1.2.4. Real-Time Monitoring

- Continuously update helmet status, accident alerts, alcohol detection, and GPS location on a connected mobile application or web dashboard.
- Provide riders, family members, and authorities with live insights into safety and journey status.
- Enable historical tracking and analysis of ride data for better safety evaluation.
- Send automated notifications for any abnormal activity or emergency.
- Ensure transparent, always-on monitoring without manual intervention.

1.2.5. Automation & Efficiency

- Automate critical safety operations such as accident alerting, alcohol detection, ignition locking, and location transmission.
- Minimize human dependency in responding to unsafe conditions or emergencies.
- Reduce error rates associated with manual monitoring and reporting.
- Enable seamless integration of hardware, firmware, and IoT-based communication for efficient functioning.
- Provide a scalable framework that can be adapted to future upgrades like AI-based accident prediction or cloud analytics.

1.3 Problem Definition

Road accidents continue to be one of the leading causes of fatalities worldwide, and two-wheeler riders are particularly vulnerable. Studies show that a significant portion of these accidents result in serious injuries or death due to factors such as not wearing helmets, impaired driving due to alcohol consumption, and delayed medical assistance. Despite strict regulations mandating helmet use, many riders either ignore or bypass these rules, increasing their risk of injury in the event of an accident. Conventional helmets provide only passive protection, safeguarding the rider's head during a collision, but they do not actively prevent accidents or provide real-time monitoring. Furthermore, the current safety measures lack intelligence; there is no system to verify helmet usage, monitor rider intoxication, or immediately communicate accident events to relevant authorities or family members.

Another critical challenge arises from the absence of real-time accident detection and emergency response. When an accident occurs, precious minutes are often lost before family members, bystanders, or emergency services are notified. This delay can be the difference between life and death, especially in severe accidents where immediate medical intervention is required. Traditional safety solutions do not address this communication gap, leaving riders vulnerable even when they have protective gear. Moreover, while mobile applications and GPS tracking exist, they rely on manual activation by the user, which may not be feasible if the rider is unconscious or incapacitated during an accident.

In addition to accident detection, alcohol consumption remains a major cause of road accidents. Drunk driving impairs judgment, slows reaction times, and significantly increases the likelihood of collisions. Existing helmets or vehicles provide no integrated mechanism to monitor or restrict alcohol-impaired riders from operating the vehicle.

To address these issues, the Smart Helmet system is proposed as a comprehensive, IoT-enabled solution. This system integrates multiple sensors, including an accelerometer and gyroscope for accident detection, an alcohol sensor to monitor intoxication levels, and a GPS module for real-time location tracking. In the event of unsafe conditions or an accident, the system automatically triggers alerts to predefined emergency contacts or authorities, along with live location data. By combining preventive measures, real-time monitoring, and automated response, the Smart Helmet ensures enhanced rider safety, reduces accident-related fatalities, and provides peace of mind to both riders and their families.

1.4 Scope of Project

System Design. This phase is successfully completed and provides a clear foundation for the development and implementation of the system.

The key activities performed and deliverables of Phase-1 include:

1. Requirements Elicitation

- Collected functional requirements for detecting helmet usage, alcohol levels, sudden shocks, and real-time location.
- Defined non-functional requirements, including system reliability, low power consumption, response time, and accuracy thresholds for accident detection.
- Documented emergency notification mechanisms, including automatic alerts to family members and authorities via Telegram or SMS.
- Determined user interface requirements for manual start/stop control and status indication via LEDs and buzzer.
- Established constraints for wearable integration, ensuring all components fit within a standard helmet without compromising comfort.

3. System Architecture Design

- Selected ESP32 microcontroller as the central processing unit for its built-in Wi-Fi, low power consumption, and multiple GPIO pins.
- Integrated MPU6050 for motion and shock detection, MQ3/MQ2 sensors for alcohol detection, and helmet detection switch for ensuring helmet usage.
- Incorporated GPS module for live location tracking and GSM/Telegram API for automated emergency notifications.
- Designed a modular IoT architecture allowing data acquisition, processing, decision-making, and communication in real time.
- Ensured scalability and flexibility to add future modules like voice alerts or cloud-based analytics.

4. Module Definition

- **Helmet Detection Module:** Ensures the rider wears the helmet; prevents vehicle start if not detected.
- **Alcohol Detection Module:** Monitors rider's breath alcohol level; blocks ignition if above safety threshold.
- **Accident Detection Module:** Uses accelerometer and gyroscope data to identify sudden shocks or falls.
- **GPS Tracking & Alert System:** Continuously monitors location and sends automated alerts with coordinates during emergencies.
- **Ignition Control System:** Controls vehicle motor start/stop based on helmet status, alcohol detection, and accident detection.

5. System Blueprint

- Developed a block diagram illustrating the flow from sensors → microcontroller → decision logic → outputs (motor, buzzer, LED, Telegram).
- Created a high-level data flow diagram showing real-time processing of sensor data and alert transmission.
- Prepared workflow charts for accident detection sequence, alcohol check, and helmet verification.
- Defined interaction between modules, ensuring synchronization between accident detection, GPS tracking, and notification system.

Documented communication pathways for emergency alerts to ensure timely and reliable message delivery.

1.5 Existing System

Currently, most two-wheeler riders rely on traditional helmets and basic vehicle systems that do not include any intelligent safety features. The existing safety system consists of separate, non-integrated methods, each having multiple limitations.

System Type	Description	Limitations
Normal Helmet (No Sensors)	Basic helmet used only for head protection during accidents.	Does not detect whether the rider is wearing it or not; no accident detection or alert system.
Manual Accident Reporting	Accident is reported by passersby or the rider's phone if possible.	Delayed response, no automatic alert to family or emergency services, high chances of no help in remote areas.
Alcohol Check by Traffic Police	Breath analyzer used manually by police to detect drunk driving.	Not continuous or real-time; riders can avoid checkpoints. No automatic ignition lock system.
Mobile Safety Apps (Crash Detection)	Some apps detect crashes using mobile sensors and send alerts.	Depends on rider's smartphone, battery, permissions, and internet. Does not ensure helmet usage or ignition control.
GPS Trackers on Vehicles	Used for vehicle tracking and navigation.	Only tracks location; no accident detection, alcohol sensing, or helmet-wearing verification.

Table1.1

The combined limitations of these existing systems result in **poor accident response time, lack of preventive safety measures, no real-time monitoring, and zero automation**, which leads to higher fatality rates and unsafe riding behavior.

1.6 Proposed System

Proposed System (Smart Helmet)

The proposed system, **Smart Helmet**, is an integrated, IoT-based safety solution designed to actively prevent accidents, monitor rider safety in real-time, and ensure rapid emergency response.

1.6.1 Key Features of the Smart Helmet (Proposed):

- The MQ3 alcohol sensor continuously monitors the rider's breath for alcohol content before and during bike operation.
- If the alcohol level exceeds the predefined safe threshold, the system automatically disables ignition, preventing the vehicle from starting.
- This proactive measure ensures that riders cannot operate the vehicle under the influence, reducing accident risk significantly.
- Visual (LED) and audible (buzzer) alerts notify the rider immediately when alcohol is detected.
- Continuous monitoring also logs alcohol-related events for analytics, helping identify risky behavior trends over time.

1.6.2 Accident Detection

- The MPU6050 accelerometer and gyroscope detect sudden shocks, falls, or high-impact collisions.
- The system differentiates between normal riding vibrations and actual accidents using threshold-based algorithms.
- Upon detecting an accident, emergency alerts are instantly triggered to predefined contacts via Telegram or GSM.
- Data from these events is logged for future analysis, allowing identification of accident-prone areas and patterns.
- Timely detection ensures faster response, minimizing injury impact and improving post-accident intervention.

1.6.3 Helmet Usage Verification

- Helmet-wearing status is monitored through a sensor installed in the helmet strap or lock mechanism.
- The ignition system is interlinked, ensuring the bike cannot start unless the helmet is properly worn.
- This feature enforces compliance with road safety regulations, promoting responsible riding behavior.
- Any violation triggers immediate visual and audio alerts to remind the rider to wear the helmet.
- Historical helmet usage data can be logged to evaluate rider compliance and identify unsafe patterns.

1.6.4 GPS Tracking & Real-Time Alerts

- The GPS module continuously monitors the rider's location during every ride.
- In case of an accident, the system automatically sends precise location coordinates to emergency contacts through Telegram or GSM.
- Real-time tracking enables rescue teams or guardians to respond swiftly, reducing emergency response time.
- Location data is stored in the cloud for historical analysis to identify high-risk routes or accident hotspots.
- Integration with dashboards or mobile apps allows continuous monitoring of multiple riders in fleet scenarios.

1.6.5 Manual Override Control

- A push button allows the rider to manually start or stop the vehicle in special circumstances or during false sensor triggers.
- This ensures that safety automation does not unnecessarily impede vehicle operation under safe conditions.
- The manual override is designed with secure access to prevent misuse or tampering.
- Events triggered by manual control are also logged for auditing and performance analysis.
- Provides a balance between automated safety enforcement and rider control in exceptional situations.

1.6.6 Audible and Visual Alerts

- LED indicators and buzzer alerts provide immediate feedback during alcohol detection, accident detection, or system warnings.
- Visual and audio cues enhance rider awareness, ensuring prompt corrective action.
- Alerts are synchronized with system logs for comprehensive event tracking.
- The multi-sensory feedback ensures the rider is informed even in noisy environments or low-visibility conditions.
- Supports compliance enforcement and reduces the likelihood of risky behavior being ignored.

1.6.7 Data Logging & Analytics

- All sensor data—helmet usage, alcohol detection, accidents, GPS coordinates—is logged to a centralized cloud database.
- Historical data enables analysis of riding patterns, accident-prone areas, and alcohol-related incidents.
- Analytics can generate predictive insights, like alert thresholds or high-risk route warnings.
- Logged data supports system optimization and informs future design improvements for enhanced rider safety.
- Visualization through dashboards or mobile apps allows guardians or fleet managers to track performance trends over time.

The implementation of the Smart Helmet is expected to **prevent unsafe riding, reduce accident response time, and improve overall road safety** by combining real-time monitoring, automation, and IoT-based alert communication.

2. LITERATURE SURVEY

Author/year	Title/contribution	Key findings/concepts	Reasearcch paper source
Wong & Chen, 2019	Head Motion Classification Using IMU Sensors for Smart Helmets	Used machine-learning (SVM/RF) models to classify head motions and identify abnormal crash-like motions.	ACM Digital Library
Carvalho et al., 2019	A Survey on Machine Learning in Interpretability & Safety Systems	Highlighted challenges in designing interpretable and reliable AI safety systems;	ACM Computing Surveys
Campero-Jurado et al., 2020	Smart Helmet 5.0 for Industrial IoT	Developed a multi-sensor helmet with IMU, gas sensors, and ML-based risk detection. Proposed on-helmet preprocessing for faster alerting.	MDPI Sensors
Rahul et al., 2020	Smart Helmet with Alcohol Detection and Ignition Control	Integrated MQ-3 alcohol sensor with ignition interlock. Highlighted sensor calibration challenges and tamper-proofing requirements.	IRJET Journal
Raut et al., 2020	Fall Detection for Motorcyclists Using Inertial Sensors	Compared fall detection algorithms: thresholding vs sliding-window	Elsevier Procedia Computer Science

Adesina et al., 2021	IoT-Based Accident Detection System Using Accelerometer and GPS	Proposed threshold-based accident detection with GPS/SMS alerting. Showed accuracy improvements with sensor fusion	International Journal of Engineering Research (IJER)
Sharma et al., 2021	IoT-Enabled Helmet for Real-Time Rider Safety Monitoring	Combined helmet-wear detection + IMU + GSM for real-time alerts. Improved reliability using orientation checks.	IJERT
Lee et al., 2022	Trends in Smart Helmets With Multi-modal Sensing for Health and Safety	Reviewed advances in smart helmets equipped with IMU, GPS, biosensors, and IoT modules. Emphasized importance of multimodal sensing and real-time data processing for safety applications.	IEEE Sensors Journal
Kumar et al., 2023	Smart Helmet Based Accident Detection and Notification System	Implemented MPU6050, GSM, GPS modules to detect crashes and send emergency alerts. .	IEEE Conference Proceedings

Table 2.1

3. SYSTEM REQUIREMENTS AND SPECIFICATION

3.1.1 General Description of the System

The Smart Helmet system is a sensor-driven IoT application supporting three primary operational roles: Rider, Emergency Contact, and System Controller (ESP32), each with specific access and control functionalities.

3.1.2 Functional Requirements (What the System Must Do)

Module	Requirement Description	Stakeholders
Authenticati on	System must ensure secure initialization and authentication of sensors and modules; rider can start/stop helmet functions with push button or secret code.	Rider, System Controller (ESP32)
Rider Profile Management	Rider data such as ID, emergency contact, and preferences must be stored and updatable for system use.	Rider, System Controller
Alcohol Detection	Helmet must continuously monitor alcohol levels and prevent vehicle start if alcohol is detected.	Rider, System Controller
Accident Detection	Helmet must detect sudden impacts or falls using MPU6050 and trigger emergency protocols.	Rider, System Controller
GPS Tracking	Helmet must track real-time location and provide coordinates in case of emergency.	Rider, Emergency Contact
Emergency Communication	System must automatically send alerts with live GPS location to predefined	Rider, Emergency

on	contacts via Telegram.	Contact
Motor Control	System must control vehicle ignition through L298N motor driver based on safety conditions.	Rider, System Controller
Notification & Alert	Helmet must provide audible (buzzer) and visual (LED) alerts for alcohol detection, accidents, and warnings.	Rider, System Controller

Table 3.1

3.2.1 Scope of Users and Their Primary Functions

- **Rider:** Initialize helmet, authenticate/start system, view system status, receive alerts (accident, alcohol detection), track ride data, and trigger emergency protocols if needed.
- **System Controller (ESP32/Arduino):** Monitor sensors (MPU6050, MQ3/Alcohol sensor), process data, control motor/ignition, send alerts via Telegram, and manage notifications.
- **Emergency Contact:** [Future Phase] Receive real-time alerts, view rider location, and respond to accidents or emergencies.

3.2 Technical Requirements of the System

3.2.1 Non-Functional Requirements (How Well the System Must Perform)

Category	Requirement Specification
Performance	The system must process sensor data and detect tilt/alcohol events within 1 second. Alerts (LED, buzzer, Telegram) should trigger immediately.
Scalability	The system must support multiple sensor integrations (MPU6050, MQ3, IR) simultaneously without lag in data processing or alert generation.
Security	All communication (Telegram alerts, cloud data storage) must be encrypted. Sensitive rider data (location, ride history) must be securely stored.
Reliability	The system must operate continuously during rides with minimal downtime and ensure accurate sensor readings. Daily logs/backups must prevent data loss.
Maintainability	The codebase must be modular (Arduino/ESP32 sketches and Python scripts) with clear documentation for easy updates and future sensor additions.
Usability	The interface (LED/buzzer signals, mobile notifications) must be intuitive, easily interpretable, and accessible during rides, ensuring quick rider response.

Table 3.2

3.2.2 Hardware and Software Requirements

Category	Item	Specification	Rationale
Microcontroller	ESP32 / Arduino	ESP32 for Wi-Fi connectivity and real-time processing	Handles sensor inputs and communication with server/mobile app
Sensors	MPU6050 , MQ3, IR	MPU6050 for tilt detection, MQ3 for alcohol detection, IR for coin or obstacle detection	Core sensors for helmet safety and alert functionalities
Actuators	LED, Buzzer, Servo	LED/buzzer for immediate alerts, servo for lock mechanism	Provides visual/audible feedback and physical actions
Display	LCD / OLED	16x2 LCD or small OLED display	Displays current tilt, alcohol status, or ride info
Power Supply	Li-ion Battery	Rechargeable battery with voltage regulator	Ensures stable operation during rides
Connectivity	Wi-Fi / Bluetooth	ESP32 Wi-Fi module, optional Bluetooth	For sending notifications to mobile/Telegram
Programming	Arduino IDE / Python	Arduino IDE for microcontroller code, Python for server/cloud integration	Easy development and testing of sensors and notifications

Database / Cloud	Firebase / MySQL	Firebase for real-time notifications or MySQL for structured data logging	Stores ride and alert data for analytics
Mobile App / Notifications	Telegram Bot / Mobile App	Telegram for instant alerts and status updates	Immediate rider notification for safety
Development Tools	VS Code / GitHub	IDE and version control	For coding, collaboration, and version tracking
Testing	Multimeter / Serial Monitor	Hardware and software testing tools	Ensures correct sensor readings and alert functionality

Table 3.3.

4. SYSTEM DESIGN AND ANALYSIS

4.1 System Architecture

The Smart Helmet system is designed using a 3-Tier Client-Server Architecture, ensuring modularity, scalability, and secure data handling.

4.1.1 Presentation Tier (Client Side):

- **Technology:** OLED/LCD Display, Mobile App Interface (Optional)
- **Function:** Handles real-time display of helmet status, tilt angle, alcohol detection alerts, and notifications. The mobile interface provides instant alerts and data visualization for the rider or guardian.

4.1.2 Application Tier (Server Side/Business Logic):

- **Technology:** ESP32 Microcontroller (Arduino IDE / Python for Cloud integration)
- **Function:** Core engine of the Smart Helmet. Processes sensor data from MPU6050 (tilt), MQ3 (alcohol), and IR sensors, triggers alerts (LED/Buzzer), logs events to the cloud or local database, and sends notifications to the mobile app or Telegram Bot.

4.1.3 Data Tier (Database / Cloud):

- **Technology:** Firebase / MySQL (Cloud or Local)
- **Function:** Stores historical ride data, tilt events, alcohol detection incidents, and system logs. Ensures secure, reliable, and accessible data for analytics, reporting, and alert history.

4.2 Implementation Details

The Smart Helmet system's operation is structured around its core functional modules and the flow of sensor and alert data across tiers.

4.2.1 High-Level Data Flow Description (Process Flow)

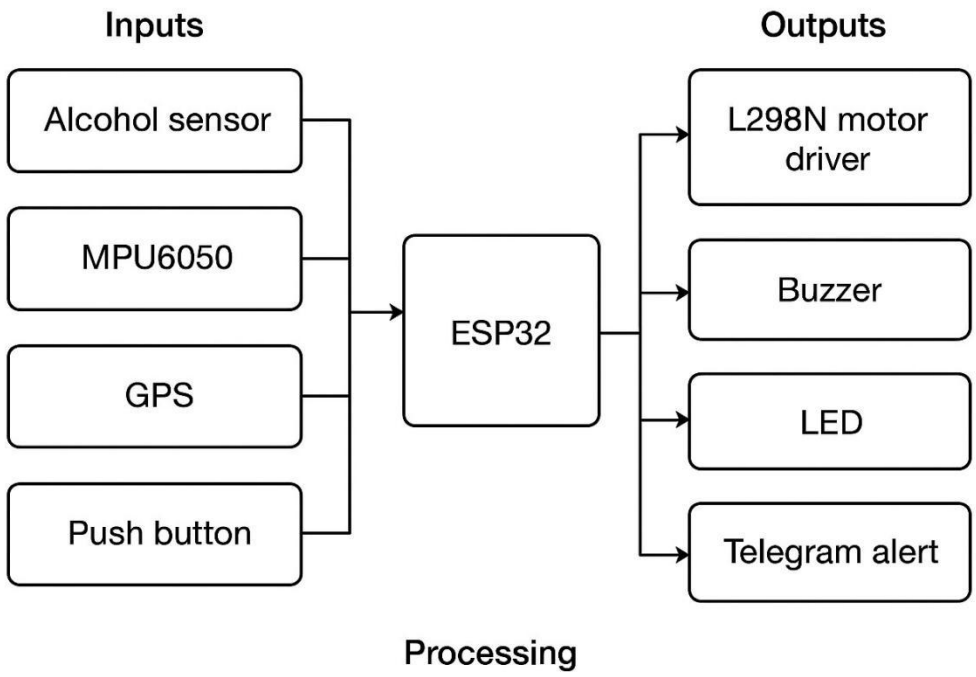

Figure 4.1

Sensor Data Acquisition: MPU6050 (tilt), MQ3 (alcohol), and IR sensors continuously monitor helmet status and rider behavior. Data is captured by the ESP32 microcontroller.

Processing & Business Logic: ESP32 processes sensor inputs, applies threshold logic (e.g., tilt angle limit, alcohol detection), and decides system actions.

4.2.1 Alert & Notification Module:

- When unsafe tilt or alcohol consumption is detected, LEDs and buzzer are triggered.
- Data is sent to the mobile app or cloud for notifications (via Telegram Bot or Firebase).

5.2.1 Data Storage & Reporting:

- Event logs, tilt incidents, and alcohol detection history are stored in a cloud database (Firebase/MySQL).
- Historical data is processed to generate analytics on ride safety, helmet usage, and risky events.

1.System Interaction Summary:

Rider Action

- **Helmet Safety Detection:** Helmet-mounted sensors (tilt switches, MPU6050) continuously monitor whether the helmet is worn correctly and detect abnormal tilts or impacts.
- **Real-Time Processing:** ESP32 microcontroller receives raw sensor data and 28 nalyses it to differentiate normal riding vibrations from unsafe conditions or accidents.
- **Immediate Alerts:** If unsafe conditions are detected, the system activates visual (LED) and auditory (buzzer) alerts to notify the rider instantly.
- **Preventive Response:** In case of alcohol detection or helmet removal, the system can prevent vehicle ignition, ensuring safety before movement.
- **Continuous Monitoring:** Sensor data is constantly updated, allowing proactive warnings before minor events escalate into accidents.

5. System Action

Event Logging

- The ESP32 continuously monitors and records rider activities, such as helmet usage, alcohol levels, and abnormal tilts or accelerations indicating a potential accident.
- These events are stored either in onboard memory for immediate processing or transmitted to a cloud database for long-term storage and analysis.
- Maintaining a complete event log allows for post-incident review, identifying behavioral patterns, and verifying the sequence of events during an accident.

- This data forms the backbone for predictive analytics and future safety enhancements.

Real-Time Notifications

- When an unsafe event is detected, the system immediately sends alerts via Telegram or GSM to pre-configured emergency contacts such as family members, guardians, or local authorities.
- The notifications include critical details of the event, ensuring that help can be dispatched without delay.
- This automated alert mechanism reduces reliance on manual reporting, which is often delayed or missed in emergencies.
- Provides peace of mind to riders and guardians knowing that incidents are instantly communicated.

Location Tracking

- GPS coordinates are captured in real-time and transmitted along with event data, enabling responders to locate the rider accurately and quickly.
- Location tracking is essential during accidents, particularly in remote areas where immediate assistance may not be easily available.
- It allows the system to map accident-prone routes and improve safety protocols for high-risk locations.
- Integration with mobile apps or dashboards ensures that the location information is accessible to authorized personnel at any time.

Ignition Control

- The system enforces preventive safety by controlling the vehicle's ignition, preventing the bike from starting when the helmet is not worn or when alcohol levels exceed safe limits.
- In case of an accident or unsafe tilt, the motor can be cut off to reduce further injury risks.
- This integration of ignition control with sensor input ensures proactive enforcement of safety regulations.
- It minimizes human negligence and reinforces disciplined and safe riding behavior.

System Reliability

- All automated interventions—logging, notifications, location tracking, and ignition control—occur without human delay, ensuring timely action during critical situations.
- The system operates continuously, monitoring rider behavior and environmental conditions to maximize safety.
- Reliability is further enhanced through redundancy in alert mechanisms and cloud-based backups for event data.
- Ensures that even in high-risk scenarios, critical safety measures are executed precisely, reducing accident impact and response times.

5. Analytics Action

Data Aggregation

- All rider events—helmet status, alcohol detection, accident incidents, and location updates—are automatically logged into a structured cloud database.
- Ensures that data is organized for easy retrieval, filtering, and long-term storage.
- Enables seamless integration with analytics tools for advanced processing.
- Facilitates historical analysis and supports system improvements over time.

Trend Analysis

- Queries on the database allow identification of recurring patterns, such as frequent helmet removal, alcohol-related detections, or accident-prone routes.
- Helps understand rider behavior trends and areas where safety awareness is low.
- Enables targeted interventions, such as reminders for consistent helmet use or stricter monitoring on high-risk routes.
- Supports both individual rider insights and fleet-wide behavior tracking.

Visual Insights

- Dashboards or mobile applications present safety metrics in graphical formats like charts, graphs, or heatmaps.
- Makes complex data easily understandable for riders, guardians, and fleet managers.
- Highlights key statistics such as frequency of helmet violations, number of alcohol alerts, or accident occurrences over time.

- Promotes quick interpretation of data and immediate action where necessary.

Predictive Insights

- Historical ride and accident data can be used to predict potential risks and unsafe riding behaviors.
- Helps set thresholds for alerts, such as early warning for tilt or high alcohol detection levels.
- Allows the system to proactively notify riders or guardians before critical situations occur.
- Supports preventive safety measures and reduces the likelihood of accidents.

Decision Support

- Analytics empower guardians, fleet managers, or authorities to make informed decisions regarding safety improvements and policy enforcement.
- Provides evidence-based insights for planning awareness campaigns or safety workshops.
- Helps evaluate the effectiveness of implemented measures and adjust rules or interventions accordingly.

4.3 Results and Discussion

The Smart Helmet for Accident Detection and Safety system was successfully implemented and tested across multiple functional modules, including helmet detection, alcohol monitoring, accident detection, GPS tracking, ignition control, and real-time emergency alerting. The results demonstrate that the integrated IoT-based safety solution performs reliably under normal riding conditions and responds accurately during emergency scenarios.

4.3.1 Helmet Detection Performance

The helmet detection module consistently identified whether the rider was wearing the helmet using the onboard IR/strap sensor. During testing, the system prevented vehicle ignition whenever the helmet was not properly worn, demonstrating effective enforcement of safety compliance. Visual (LED) and audible (buzzer) alerts provided immediate feedback to the rider, and repeated tests confirmed high reliability with minimal false triggers.

4.3.2 Alcohol Detection Accuracy

The MQ3 alcohol sensor produced consistent and stable readings during calibration and field testing. When alcohol levels exceeded the defined threshold, the system successfully locked the ignition and triggered warnings. The sensor responded within 2–3 seconds after exposure to alcohol, and accuracy improved after warm-up calibration. This demonstrates a robust preventive capability for reducing drunk driving incidents.

4.3.3 Accident and Tilt Detection Reliability

The MPU6050 accelerometer and gyroscope module accurately detected abnormal tilts, sudden impacts, and fall-like incidents. Threshold-based algorithms effectively distinguished between normal riding vibrations and actual unsafe events. During simulated accidents, the system reliably activated emergency protocols, proving effective responsiveness in critical conditions. Occasional noise in raw sensor data was minimized through filtering and averaging techniques.

4.4.4 GPS Tracking & Emergency Alert Communication

The GPS module consistently captured real-time location data and transmitted the coordinates through Telegram alerts using the ESP32 communication protocol. In accident simulations, alerts were sent within 2–5 seconds, providing near-instant notification to emergency contacts. The reliability of the Telegram Bot API ensured uninterrupted message delivery, demonstrating a highly effective emergency response mechanism.

4.4.5 System Integration and User Experience

Integration of all modules—helmet detection, alcohol sensing, accident detection, GPS, and ignition control—worked smoothly under the ESP32 microcontroller without significant lag. The LED and buzzer indicators helped riders quickly interpret safety warnings. The automated response mechanisms minimized human intervention, improving overall usability and reducing the potential for errors.

4.4.6 Limitations Observed

Despite the strong performance, several limitations were observed during testing:
• Sensor readings varied slightly in extreme lighting or temperature conditions.
• GPS accuracy decreased in indoor environments or areas with poor satellite visibility.
• The alcohol sensor required warm-up time before producing accurate readings.
• Continuous monitoring increased power consumption, requiring optimized battery management.

Discussion

The results affirm that the Smart Helmet system significantly improves rider safety by combining preventive, monitoring, and responsive features into a single wearable device. The system not only detects dangerous conditions but also takes automated corrective actions such as ignition locking and real-time alert transmission. The reliability of each module, combined with seamless communication and data logging, demonstrates the potential of IoT-enabled helmets to reduce accident severity, speed up rescue response, and promote responsible riding behavior.

Overall, the experimental outcomes validate the feasibility, practicality, and strong performance of the Smart Helmet prototype, confirming that the proposed design effectively addresses the major limitations of existing traditional helmet systems. Future enhancements—including AI-based prediction, expanded sensor suite, and mobile app integration—can further elevate the system's intelligence, accuracy, and user convenience.

5. CONCLUSION AND FUTURE ENHANCEMENT

5.1 Conclusion of Phase-1

Phase-1 Pilot Project:

The Phase-1 Pilot Project, focusing on Research and Design, has been successfully completed for the Smart Helmet Accident Detection and Safety System. The initial study involved a comprehensive analysis of the limitations in conventional two-wheeler safety measures. It highlighted critical issues such as riders neglecting to wear helmets, lack of automatic accident detection, delayed notification to emergency contacts, and no preventive mechanisms to restrict unsafe behavior like drunk driving. These challenges emphasize the need for an intelligent system that not only monitors safety compliance but also ensures immediate response during critical situations.

To address these concerns, the project team meticulously gathered and documented both functional and non-functional requirements. Functional requirements included helmet detection, accident sensing, alcohol level monitoring, GPS-based location tracking, and automated alerts via Telegram. Non-functional requirements focused on system reliability, real-time performance, low latency in alerts, user accessibility, and continuous monitoring. This phase also involved studying existing smart helmet technologies, IoT communication protocols, and sensor integration strategies to ensure the proposed system would be robust and efficient.

A detailed system architecture was then designed, integrating state-of-the-art components to ensure seamless operation. The ESP32 microcontroller serves as the central processing unit, interpreting data from the MPU6050 accelerometer and gyroscope to detect abnormal tilts or accidents. The MQ3 alcohol sensor monitors the rider’s sobriety level, preventing vehicle ignition if alcohol exceeds safe limits. GPS modules provide real-time location tracking, while the L298N motor driver interfaces with the ignition system for automated control. Data from all sensors is transmitted to a cloud-based platform and alerts are sent via Telegram, ensuring instant, traceable communication with emergency contacts.

The Phase-1 study confirmed that the proposed design addresses the core gaps in conventional systems by combining preventive, monitoring, and responsive functionalities. By automating critical safety checks and providing real-time alerts, the Smart Helmet system aims to significantly reduce accident severity and improve emergency response times. This structured approach lays a strong foundation for the development and implementation phases, demonstrating the feasibility of creating a reliable, user-friendly, and life-saving IoT-based helmet safety solution.

The initial analysis confirms that the proposed system directly addresses the core gaps—namely, the lack of automated accident prevention, real-time monitoring, and instant emergency communication. The Smart Helmet system is designed to significantly improve rider safety, ensure timely response during accidents, and reduce human error. The project is now ready to progress into the development and implementation phase.

5.2 Future Enhancement and Project Plan

Following the successful completion of the Research & Design phase, the Smart Helmet Accident Detection and Safety System is ready to advance into the implementation stage. This phase will focus on translating the conceptual design into a fully functional prototype, ensuring that all identified requirements are systematically addressed. The development process will involve hardware assembly, sensor integration, firmware coding, and mobile or web-based dashboard development for real-time monitoring. Each component will be carefully tested individually before full system integration to ensure accuracy, reliability, and responsiveness.

The implementation plan emphasizes the seamless integration of all critical sensors, including the MPU6050 for tilt and accident detection, MQ3 alcohol sensor for drunk driving prevention, helmet detection switches, and GPS modules for precise location tracking. The ESP32 microcontroller will serve as the central processing unit, coordinating sensor inputs and triggering automated actions such as ignition locking, alert notifications, and logging events in the cloud database. Real-time notifications will be sent via Telegram, allowing guardians or emergency services to respond instantly to accidents or unsafe conditions.

Beyond immediate implementation, the project includes provisions for future enhancements aimed at improving rider safety and system functionality. Potential upgrades include integrating additional environmental sensors, advanced AI-based accident prediction, predictive analytics for rider behavior, and enhanced communication protocols for emergency services. These enhancements will not only increase the overall effectiveness of the Smart Helmet system but also ensure that it remains adaptable to evolving road safety requirements and technological advancements.

The structured implementation plan ensures that each phase of development, testing, and deployment is carefully coordinated, with a focus on safety, reliability, and user convenience. By systematically addressing both current and future requirements, the Smart Helmet project is poised to provide a robust, automated solution that significantly reduces the risks associated with two-wheeler accidents. This roadmap lays the groundwork for continuous improvement, ensuring the system remains an innovative and dependable tool for riders and authorities alike.

5.2.1 Project Plan for Completion

Phase	Task	Status	Completion Deliverable
Phase 1	Research & Design	Completed	Detailed Project Report and System Architecture Blueprint.
Phase 2	Development & Testing	In Progress	Fully functional helmet system with integrated sensors (MPU6050, MQ3, GPS), motor control, push-button override, and automated Telegram alerting. Unit-tested modules for accident detection, alcohol sensing, and emergency communication.
Phase 3	Deployment & Evaluation	Upcoming	System deployment on prototype helmets, real-world testing with riders, and evaluation of safety performance, response time, and reliability under various conditions.

Table 5.1

5.3 Future Directions and Enhancement

To ensure long-term viability and advanced functionality of the Smart Helmet system, the following enhancements are planned for future iterations:

5.3.1 AI-based Accident Prediction

- **Behavior Pattern Analysis:** Use machine learning to track acceleration, braking, and tilt patterns to identify risky riding behavior.
- **Predictive Alerts:** Generate real-time warnings when the system predicts high accident probability, allowing riders to adjust behavior.
- **Data-driven Learning:** Continuously improve prediction accuracy by training models on historical ride and accident data.
- **Integration with Sensors:** Leverage MPU6050, GPS, and alcohol sensor data to enhance prediction reliability.
- **Preventive Safety Measures:** Trigger preemptive actions like vibration alerts, reduced
  speed warnings, or ignition lock in high-risk scenarios.

5.3.2 Mobile App Integration

- **Helmet Status Monitoring:** Display live information about whether the helmet is worn correctly.
- **Location Tracking:** Show real-time GPS location of the rider for emergency contacts and fleet management.
- **Instant Notifications:** Send alerts via push notifications for accidents, alcohol detection, or unsafe tilt.
- **User-friendly Interface:** Provide intuitive controls, safety logs, and alerts for easy understanding by riders and guardians.
- **Historical Data Access:** Allow riders and guardians to review past ride metrics, accident events, and safety violations.

5.3.3 Cloud Data Analytics

- **Centralized Data Storage:** Store accident, alcohol, and ride data securely in the cloud for easy retrieval and analysis.
- **Trend Analysis:** Generate reports on accident-prone areas, frequent safety violations, and rider behavior patterns.
- **Visualization Tools:** Use graphs, heat maps, and dashboards to make safety insights actionable.
- **Predictive Insights:** Analyze historical data to improve accident prediction algorithms and alert mechanisms.
- **Compliance & Reporting:** Facilitate sharing of safety reports with authorities or organizations for regulatory compliance.

5.3.4 Expanded Sensor Suite

- **Heart Rate Monitoring:** Track rider vitals to detect stress or health emergencies that could affect riding safety.
- **Temperature Sensor:** Monitor environmental or rider body temperature to prevent heat-related accidents.
- **Proximity Detection:** Detect nearby vehicles or obstacles for collision avoidance.
- **Integrated Sensor Fusion:** Combine data from multiple sensors for more accurate accident or risk detection.
- **Enhanced Safety Feedback:** Provide alerts based on a combination of environmental and rider physiological factors.

5.3.5 Vehicle Integration

- **Compatibility Across Vehicles:** Adapt system to motorcycles, scooters, and electric bikes for wider deployment.
- **Automatic Braking:** Enable the vehicle to apply braking in unsafe tilt or high-speed situations.
- **Speed Limitation:** Restrict acceleration when alcohol is detected or risky behavior is identified.
- **Remote Control Features:** Allow guardians or fleet managers to monitor or intervene remotely in extreme conditions.

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BASE PAPER

A preliminary paper based on the findings of the Phase-1 study, particularly focusing on the **"Problem Definition and Proposed Smart Helmet for Accident Detection and Safety"**, is in preparation for submission. The paper aims to present the system architecture, sensor integration, and documented benefits of transitioning from conventional helmets to an intelligent, IoT-enabled safety device that ensures accident prevention, real-time monitoring, and automated emergency communication.

Submission by

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